

Software Requirements Specification (SRS)

Project: **Blood Bank Management System**

Version: 1.0

Authors: Student Team

Date: 06-09-2026

Status: Draft

Revision History

Version	Date	Author	Change Summary
1.0	06-09-2026	Student Team	Initial SRS prepared for the Blood Bank Management System

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall Description
3. External Interface Requirements
4. System Features (Detailed)
5. Non-Functional Requirements
6. Quality Attributes & Acceptance Tests
7. UML Use-Case Diagrams
8. Requirements Traceability Matrix (RTM)
9. Responsibilities of Each Member

1. Introduction

1.1 Purpose

This document describes the requirements of the Blood Bank Management System. The system is intended to provide a simple way to manage blood donors, patients, blood donations, blood requests, approvals, and blood stock. It serves as a common reference for the development and testing team and for evaluating whether the final system meets the expected requirements.

1.2 Scope

The system covers user registration and authentication, donor and patient profile management, blood donation records, blood requests, approval or rejection by an administrator, blood inventory updates, request and donation history, and administrative dashboard information. The system focuses on management of records and availability of blood units. It does not cover laboratory testing of donated blood, physical blood collection, transportation, or clinical decision-making.

1.3 Audience

The intended audience includes student developers, testers, course instructors, system administrators, and users involved in operating the blood bank.

1.4 Definitions and Acronyms

Term	Definition
Admin	User who manages donors, patients, donations, requests, and blood stock.
Donor	Registered user who can provide blood and maintain donation records.
Patient	Registered user who can submit a request for required blood.
Blood Unit	A recorded unit of donated blood associated with a blood group and inventory status.
Blood Request	A request submitted for a blood group and required number of units.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
SRS	Software Requirements Specification.

2. Overall Description

2.1 Product Perspective

The Blood Bank Management System is a web-based application developed using Python with Django and a JavaScript-enabled web interface. The application uses a database to store user information, donor and patient details, donation records, blood requests, and inventory information. The administrator acts as the main controller of approvals and stock management.

2.2 Major Product Functions

- User registration and login.
- Role-based access for administrators, donors, and patients.
- Donor profile and donation management.
- Patient profile and blood-request management.
- Admin approval or rejection of donations and blood requests.
- Blood inventory and stock management by blood group.
- Automatic stock increase for approved donations.

- Stock deduction when an approved blood request is fulfilled.
- Viewing donation and blood-request history.
- Dashboard information for administrative monitoring.
- Basic tracking of blood-unit expiry information where recorded.

2.3 User Roles and Characteristics

- Administrator: Expected to understand basic web applications and is responsible for managing records, approvals, and inventory.
- Donor: A normal registered user who provides donation information and can view the status and history of donations.
- Patient: A normal registered user who submits blood requests and checks their status and history.

2.4 Operating Environment

The system is intended to run as a Django web application on a computer or server with Python and a supported database. Users access the system through a modern web browser over a local or network connection.

2.5 Constraints

- The system depends on the availability of the application server and database.
- Only authorized administrators should perform approval and inventory-management operations.
- Blood availability shown by the system depends on the accuracy of the recorded inventory.
- The system is a management application and does not replace medical or laboratory procedures.
- The implementation will follow the selected Django project structure and available development resources.

3. External Interface Requirements

3.1 User Interface

The system shall provide browser-based pages for registration, login, dashboards, donor and patient operations, blood requests, donation records, and administrative management. Forms should use clear labels and provide understandable validation or status messages.

3.2 Hardware Interface

No specialized hardware is required. The system runs on standard computers, laptops, or servers with network connectivity and can be accessed through a web browser.

3.3 Software Interfaces

- Django application framework for server-side processing.
- Database system for persistent storage of application records.
- HTML/CSS/JavaScript and browser technologies for the user interface.

3.4 Communications

The system shall communicate through standard HTTP/HTTPS web requests. When deployed beyond a local development environment, HTTPS should be used to protect user credentials and other sensitive information.

4. System Features (Detailed)

The following functional requirements describe the expected behavior of the Blood Bank Management System. Each requirement is written as a 'shall' statement so that it can be verified during testing.

Req ID	Requirement	Requirement (shall...)	Type	Priority	Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
BB-F-001	User Registration	The system shall allow a new donor or patient to create an account by providing the required registration information.	Functional	High	Donor / Patient	Valid registration data creates an account and invalid or incomplete data is rejected.	TC-REG-01
BB-F-002	User Login	The system shall authenticate registered users using their login credentials.	Functional	High	Donor / Patient / Admin	A user with valid credentials is logged in and an invalid login is rejected.	TC-AUTH-01
BB-F-003	Role-Based Access	The system shall provide access to functions according to the user's role.	Functional	High	Admin / All users	A donor or patient cannot access administrator-only pages or actions.	TC-AUTH-02
BB-F-004	Donor Profile	The system shall allow donors to view and update their profile information.	Functional	Medium	Donor	Updated donor information is displayed after successful saving.	TC-DON-01
BB-F-005	Donation Submission	The system shall allow a donor to submit a blood donation record with the required donation details.	Functional	High	Donor	A valid donation submission is stored with a pending status.	TC-DON-02
BB-F-006	Donation Review	The system shall allow an administrator to approve or reject a submitted donation.	Functional	High	Admin	The selected donation status changes to Approved or Rejected and is visible in history.	TC-DON-03

BB-F-007	Donation History	The system shall allow a donor to view the status and history of their donation records.	Functional	Medium	Donor	The donor can see previous donations and their current statuses.	TC-DON-04
BB-F-008	Patient Profile	The system shall allow a patient to view and update their profile information.	Functional	Medium	Patient	Updated patient information is stored and displayed correctly.	TC-PAT-01
BB-F-009	Blood Request	The system shall allow a patient to submit a request specifying the required blood group and number of units.	Functional	High	Patient	A valid request is stored with an initial pending status.	TC-REQ-01
BB-F-010	Request Review	The system shall allow an administrator to approve or reject a blood request.	Functional	High	Admin	The request status is updated and the decision is visible to the requester.	TC-REQ-02
BB-F-011	Blood Availability	The system shall allow authorized users to view available blood stock by blood group.	Functional	High	Admin / Patient	The system displays the currently recorded available quantity for the selected blood group.	TC-INV-01
BB-F-012	Inventory Update	The system shall allow the administrator to update blood stock records when required.	Functional	High	Admin	An authorized inventory update is saved and reflected in the stock view.	TC-INV-02
BB-F-013	Donation Stock Update	The system shall increase the relevant blood stock when a donation is approved by an administrator.	Functional	High	Admin	Approving a donation increases the corresponding blood-group stock by the recorded units.	TC-INV-03
BB-F-014	Request Stock Deduction	The system shall reduce the relevant	Functional	High	Admin	The stock is reduced by the issued	TC-INV-04

		blood stock when an approved request is fulfilled.				units and shall not become negative.	
BB-F-015	Request History	The system shall allow patients to view their previous blood requests and their statuses.	Functional	Medium	Patient	The patient's request history is displayed with status information.	TC-REQ-03
BB-F-016	Admin Dashboard	The system shall provide an administrator dashboard showing relevant donor, patient, request, donation, and stock information.	Functional	Medium	Admin	Dashboard values match the corresponding stored records.	TC-ADM-01
BB-F-017	User Management	The system shall allow an administrator to view and manage registered donor and patient records.	Functional	Medium	Admin	The administrator can view the registered user records and perform permitted management actions.	TC-ADM-02
BB-F-018	Expiry Tracking	The system shall store and display expiry information for blood units where such information is recorded.	Functional	Medium	Admin	Recorded expiry information can be viewed with the associated blood stock record.	TC-INV-05

5. Non-Functional Requirements (Detailed)

Req ID	Requirement Category	Requirement	Priority	Acceptance Criteria / Measurement	Test Case
BB-NF-001	Performance	Normal page requests and routine database operations should respond within 3 seconds under	High	95% of measured normal requests complete within 3 seconds.	TC-PERF-01

		normal college-project load.			
BB-NF-002	Availability / Reliability	The system shall preserve committed records and handle application errors without silently losing submitted donation or request data.	High	Tested submissions remain available after successful save and controlled restart.	TC-REL-01
BB-NF-003	Security	Passwords shall not be stored in plain text, and authentication shall use the security mechanisms provided by Django.	High	Database inspection and authentication tests confirm password protection.	TC-SEC-01
BB-NF-004	Usability	The system shall provide clear navigation, readable forms, and understandable success/error messages for common operations.	Medium	Representative users complete registration, donation, and request tasks without unclear system messages.	TC-UX-01
BB-NF-005	Maintainability	The application shall be organized into logical Django modules with clear separation between user, donation, request, and inventory functionality.	Medium	Code review confirms modules can be modified without unnecessary changes to unrelated features.	TC-MAINT-01
BB-NF-006	Data Integrity	The system shall prevent blood inventory quantities from becoming negative and shall validate required fields before saving records.	High	Invalid quantities and incomplete forms are rejected; stock never falls below zero.	TC-DATA-01

5.1 Security

5.1.1 Security Objectives

- Protect user accounts and personal information from unauthorized access.
- Ensure that only authorized administrators can approve donations, approve blood requests, and modify blood inventory.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case
BB-SR-001	The system shall use Django's password hashing mechanisms and shall not store user passwords in plain text.	Security	High	Database/authentication review; TC-SEC-01
BB-SR-002	The system shall require authenticated access for donor, patient, and administrator functions that contain protected records.	Security	High	Unauthenticated requests are redirected or denied; TC-SEC-02
BB-SR-003	The system shall enforce role-based authorization so that administrator-only operations cannot be performed by donors or patients.	Security	High	Unauthorized role access is denied; TC-SEC-03
BB-SR-004	The system shall validate and sanitize user-submitted form data to reduce invalid input and common web attacks.	Security	High	Invalid/malicious test inputs are rejected or safely handled; TC-SEC-04
BB-SR-005	When deployed over a network, the system shall use HTTPS to protect credentials and sensitive user information during transmission.	Security	High	Network/deployment configuration confirms HTTPS; TC-SEC-05

6. Quality Attributes & Acceptance Tests

The system will be considered ready for submission when the major high-priority functional requirements are implemented and verified, no critical security or data-integrity issue remains open, and the RTM shows that the required test cases have been executed.

- Registration and authentication tests.
- Role and authorization tests.
- Donation submission and approval tests.
- Blood request and approval tests.
- Inventory update and consistency tests.
- Performance and usability checks.
- Security and input-validation checks.

7. UML Use-Case Diagrams

7.1 Use-Case Diagram 1 – Donor and Patient

Actors: Donor, Patient. Main use cases: Register, Login, Manage Profile, Submit Donation, View Donation History, Submit Blood Request, View Request History, Check Blood Availability.



7.2 Use-Case Diagram 2 – Administrator

Actor: Administrator. Main use cases: Login, Manage Donors, Manage Patients, Review Donations, Approve/Reject Donation, Review Blood Requests, Approve/Reject Request, Manage Blood Inventory, View Dashboard, View History, Manage Users.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)
BB-F-001	Registration	4 / User Management	User/Auth Module	TC-REG-01	N
BB-F-002	Login	4 / Authentication	Auth Module	TC-AUTH-01	N
BB-F-005	Donation submission	4 / Donation	Donor Module	TC-DON-02	N
BB-F-006	Donation approval	4 / Donation Review	Admin Module	TC-DON-03	N
BB-F-009	Blood request	4 / Blood Request	Request Module	TC-REQ-01	N
BB-F-010	Request approval	4 / Request Review	Admin Module	TC-REQ-02	N
BB-F-011	Blood availability	4 / Inventory	Inventory Module	TC-INV-01	N
BB-F-013	Donation stock update	4 / Inventory	Inventory Module	TC-INV-03	N

BB-F-014	Request stock deduction	4 / Inventory	Inventory Module	TC-INV-04	N
BB-NF-001	Performance	5 / Performance	Web App / Database	TC-PERF-01	N
BB-NF-003	Security	5.1 / Security	Auth Module	TC-SEC-01	N
BB-NF-006	Data integrity	5 / Data Integrity	Inventory Module	TC-DATA-01	N
BB-SR-003	Role authorization	5.1 / Security	Auth/Admin Module	TC-SEC-03	N

👤 Member 1 (KHUSH BHARDWAJ, PES2UG24CS226) — Introduction + Overall Description

Responsible for:

1. Introduction

- 1.1 Purpose
- 1.2 Scope
- 1.3 Audience
- 1.4 Definitions & Acronyms

2. Overall Description

- 2.1 Product Perspective
- 2.2 Major Product Functions
- 2.3 User Roles and Characteristics
- 2.4 Operating Environment
- 2.5 Constraints

Also prepare:

- Project title/version information
- Revision history
- Approval section

👤 Member 2 (QAMAR AHMED, PES2UG24CS904) — External Interfaces + Donor/Patient Requirements

Responsible for:

3. External Interface Requirements

- 3.1 User Interface
- 3.2 Hardware Interface

- 3.3 Software Interface
- 3.4 Communications

4. System Features — Donor & Patient

Take these requirements:

- BB-F-001 — User Registration
- BB-F-002 — User Login
- BB-F-003 — Role-Based Access
- BB-F-004 — Donor Profile
- BB-F-005 — Donation Submission
- BB-F-007 — Donation History
- BB-F-008 — Patient Profile
- BB-F-009 — Blood Request
- BB-F-011 — Blood Availability
- BB-F-015 — Request History

 **Member 3 (Hemaksh Breja, PES2UG24CS191) — Admin + Inventory Requirements + NFR**

Responsible for:

4. System Features — Admin/Inventory

- BB-F-006 — Donation Review
- BB-F-010 — Request Review
- BB-F-012 — Inventory Update
- BB-F-013 — Donation Stock Update
- BB-F-014 — Request Stock Deduction
- BB-F-016 — Admin Dashboard
- BB-F-017 — User Management
- BB-F-018 — Expiry Tracking

5. Non-Functional Requirements

- BB-NF-001 — Performance
- BB-NF-002 — Reliability
- BB-NF-003 — Security
- BB-NF-004 — Usability
- BB-NF-005 — Maintainability
- BB-NF-006 — Data Integrity

 **Member 4 (KOUSHIK C, PES2UG24CS232) — Security + UML + RTM**

Responsible for:

5.1 Security

- 5.1.1 Security Objectives
- 5.1.2 Security Requirements
- BB-SR-001 → BB-SR-005

6. Quality Attributes & Acceptance Tests

7. UML Use-Case Diagrams

- Donor + Patient diagram
- Administrator diagram

8. Requirements Traceability Matrix